PHD22 Macro labour: Problem set

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Question 1: The DMP model

I follow the exposition from Hagedorn and Manovskii (2008) wherever possible. The Bellman equations that govern the model are:

$$J(p) = p - w(p) + \beta(1-s)\mathbb{E}_p[J(p')], \tag{1a}$$

$$V(p) = -c + \beta q[\theta(p)]\mathbb{E}_p[J(p')], \tag{1b}$$

$$U(p) = b + \beta\langle f[\theta(p)]\mathbb{E}_p[W(p')] + \{1 - f[\theta(p)]\}\mathbb{E}_p[U(p')]\rangle, \tag{1c}$$

$$W(p) = w(p) + \beta\{(1-s)\mathbb{E}_p[W(p')] + s\,\mathbb{E}_p[U(p')]\}. \tag{1d}$$

I use the matching function from Den Haan et al. (2000) :

$$m(u,v) = \frac{uv}{(u^l + v^l)^{\frac{1}{l}}}, \tag{2}$$

which gives the following contact rates:

$$q[\theta(p)] = \left[1 + \theta(p)^l\right]^{-\frac{1}{l}}, \tag{3a}$$

$$f[\theta(p)] = \left[1 + \theta(p)^{-l}\right]^{-\frac{1}{l}}. \tag{3b}$$

Free entry condition

The **free entry condition** stipulates $V(p) = 0$ which translates to:

$$c = \beta q[\theta(p)]\mathbb{E}_p[J(p')], \tag{4a}$$

$$J(p) = p - w(p) + \frac{(1-s)c}{q[\theta(p)]}. \tag{4b}$$

The free entry condition can be used to derive the job posting condition. In the steady state, where we have only one price, $p = p^\star$, the conditions collapse to:

$$c = \beta q(\theta^\star) J^\star \implies J^\star = \frac{c}{\beta q(\theta^\star)} \tag{5a}$$

$$J^\star = p^\star - w^\star + \beta(1-s)J^\star \implies J^\star = \frac{p^\star - w^\star}{1 - \beta(1-s)}. \tag{5b}$$

Steady state job creation condition

Setting $r \equiv \frac{1}{\beta} - 1$, the **job creation condition** becomes:

$$w^\star = p^\star - \frac{r+s}{q(\theta^\star)} c. \tag{6}$$

Nash bargaining and surplus sharing

A worker and a firm split the surplus, with workers taking γ:

$$W(p) = \gamma \{J(p) + W(p) - U(p)\}, \tag{7a}$$

$$J(p) = (1-\gamma)\{J(p) + W(p) - U(p)\}. \tag{7b}$$

Using the surplus sharing condition, I solve for the closed-form solution of $w(p)$:

$$[W]: \quad W(p) - \beta \mathbb{E}_p[W(p')] = w(p) - \beta s \mathbb{E}_p[W(p') - U(p')], \tag{8a}$$

$$[U]: \quad U(p) - \beta \mathbb{E}_p[U(p')] = b + \beta f[\theta(p)] \mathbb{E}_p[W(p') - U(p')], \tag{8b}$$

which can be combined:

$$W(p) - U(p) = w(p) - b + \beta \{1 - s - f[\theta(p)]\} \mathbb{E}_p[W(p') - U(p')]. \tag{8c}$$

Now, by Nash bargaining and free entry, we must have that:

$$W(p) - U(p) = \frac{\gamma}{1-\gamma} J(p) = \frac{\gamma}{1-\gamma}\left[p - w(p) + \frac{(1-s)c}{q[\theta(p)]}\right], \tag{8d}$$

$$\mathbb{E}_p[W(p') - U(p')] = \frac{\gamma}{1-\gamma} \mathbb{E}_p[J(p)] = \frac{\gamma}{1-\gamma} \frac{c}{\beta q[\theta(p)]}. \tag{8e}$$

The last three conditions can be combined and rearranged to solve for the wage.

Equilibrium wage

By combining the free entry condition with Nash bargaining, I obtain the closed form wage:

$$w(p) = \gamma[p + \theta(p) c] + (1-\gamma) b. \tag{9}$$

Beveridge curve

From the law of motion of employment:

$$n' = (1-s) n + m(v, u) \tag{10a}$$

I obtain the steady state Beveridge curve:

$$u^\star = \frac{s}{f+s}. \tag{10b}$$

Question 2: The steady state algorithm

Algorithm 1 sets out the standard procedure of calibrating a parameter, in this case c, in the DMP model.

Algorithm 1: Calibration of vacancy posting cost to target unemployment

Input: Parameters $(l, \gamma, b, \beta, s, p^\star)$, target unemployment rate $u^\star$
Output: Calibrated $c^\star$, equilibrium objects $(\theta^\star, w^\star, q^\star, f^\star, u^\star)$

```
repeat
    // Inner steady-state problem given c
    Use the bisection (or alternative) algorithm;
    repeat
        Compute implied tightness θ from Equation (3a) ;
        Compute implied wage w from Equation (9) ;
        Compute θ̃ from the job-creation condition, Equation (6) ;
        Update q by bisection on the residual |q^{n+1} − q^n|;
    until |q^{n+1} − q^n| < ε_q;
    // Evaluate calibration target
    Compute job-finding rate f from the matching function in Equation (3b);
    Compute steady-state unemployment u from Equation (10b);
    Update c by bisection on the residual u − u*;
until |u − u*| < ε_κ;
```

However, I compute c directly, as the model allows to construct c as a function of u and remaining parameters. Start by combining the job creation condition and the equilibrium wage from Equations (6) and (9):

$$\gamma\left(p^\star + \theta^\star c\right) + \left(1-\gamma\right) b = p^\star - \frac{r+s}{q\left(\theta^\star\right)} c \tag{11a}$$

and rearrange a little bit:

$$c = \frac{\left(1-\gamma\right) b}{\frac{r+s}{q(\theta^\star)} + \gamma\theta^\star}. \tag{11b}$$

$\theta^\star$ and $q\left(\theta^\star\right)$ can be obtained from the Beveridge curve in Equation (10b):

$$u = \frac{s}{f\left(\theta^\star\right) + s}. \tag{11c}$$

"Algorithm" 2 fleshes out the fast way to do it (I abstract from computing a large closed form solution).

Algorithm 2: Calibration of vacancy posting cost (closed form)

Input: Parameters $(l, \gamma, b, \beta, s, p^\star)$, target unemployment rate $u^\star$
Output: Calibrated $c^\star$, equilibrium objects $(\theta^\star, w^\star, q^\star, f^\star)$

```
// Pin down the flow values and tightness
Compute job-finding rate f* from the Beveridge curve, Equation (10b);
Compute tightness θ* by inverting Equation (3b);
Compute job-filling rate q* from Equation (3a);
// Free entry and Nash bargaining pin down c
Jointly solve Equation (6) and Equation (9) for c*;
Compute equilibrium wage w* from Equation (9);
```

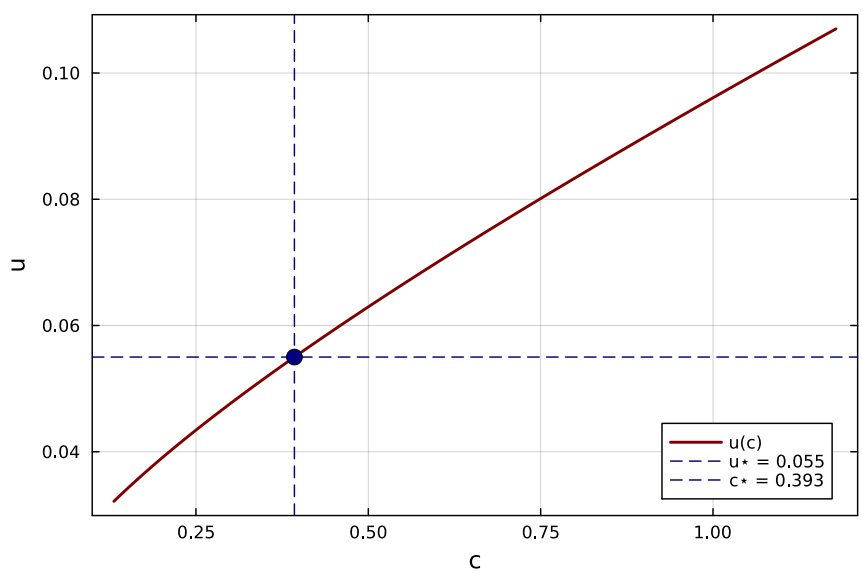
Question 3: Calibrating c

Table 1 summarises the calibration exercise. The calibrated vacancy cost is relatively high, at around 39% of the productivity value. Figure 1 indicates that the parameter is well-identified.

Table 1: Steady-state calibration

Description	**Symbol**	**Value**
Parameters		
Home production	b	0.4000
Discount factor	β	0.9992
Separation rate	s	0.0081
Bargaining power	γ	0.7000
Matching elasticity	l	0.4070
Steady-state productivity	$p^\star$	1.0000
Calibration		
Target unemployment rate	$u^\star$	0.0550
Calibrated vacancy cost	$c^\star$	0.3928
Equilibrium objects		
Labour market tightness	$\theta^\star$	0.5997
Wage	$w^\star$	0.9849
Job-filling rate	$q^\star$	0.2321
Job-finding rate	$f^\star$	0.1392
Unemployment rate	$u^\star$	0.0550
Firm value	$J^\star$	1.6938

Figure 1: Matching c to u



Question 4: Solving the model with aggregate uncertainty

I use the Rouwenhorst method to discretise the productivity process. This is more fitting for the highly persistent process, with $\rho \simeq 0.99$, than the Tauchen method. I set 15 as the number of grid points. Table 2 summarises the values of all equilibrium objects at the minimum and maximum productivity grids. Predictably, the job-finding rate varies quite little with the productivity process.

Table 2: Equilibrium objects across productivity states

Description	**Symbol**	**SS**	**Min** p	**Max** p
Productivity	p	1.0000	0.9874	1.0128
Labour market tightness	θ	0.5997	0.5870	0.6125
Wage	w	0.9849	0.9725	0.9974
Job-filling rate	q	0.2321	0.2343	0.2299
Job-finding rate	f	0.1392	0.1375	0.1408
Firm value	J	1.6938	1.6776	1.7101

Despite the rather low value of b, a worker still obtains a relatively high wage, almost as high as the productivity level. This is attributable to the large value of γ, which bolsters the worker's negotiating power. As this is the case for a low and high productivity alike, the model does not generate large enough swings in the firm value to push vacancy posting enough. This means that we cannot generate much variation in unemployment either, as one can see in the next question.

Question 5: Towards Hagedorn and Manovskii (2008)

Hagedorn and Manovskii (2008)'s Table 4 presents simulation statistics based on the simulation of monthy data, detreneded using the HP filter with a smoothing parameter of $1,600$.[1] My replication is summarised in Table 3 My results are based on a 100 year simulation ($T = 5200$ periods, with a 20-year burnin).

Table 3: Results from the calibrated model

	u	v	v/u	p
Standard deviation	0.003	0.003	0.006	0.004
Autocorrelation	1.000	0.999	1.000	1.000
Correlation matrix				
u	1.000	-0.946	-0.987	-0.987
v	—	1.000	0.986	0.986
v/u	—	—	1.000	1.000
p	—	—	—	1.000

Naturally, I expect the values to depend on three types of variables related to discretisation:

- (ρ, σ): If one adjusts the productivity process, the business cycle properties change (more below).
- T: Setting a too small value of T may mean that we don't visit far ends of each tail.
- N_p: A coarse grid may mean that we fail to capture the correlation structure of the key labour variables. Too many grid points means that the tails are too rarely visited (unless T is adjusted).

[1] Am I understading it right? I thought a value of $1,600$ was reserved for the quarterly business cycle variables.

Changing ρ even a little bit can matter a lot for volatility and the correlation structure of the key labour market variables. As illustrated in Figure 2, lowering ρ suppresses volatility in other variables. Figure 3 fleshes out the sensitivity of the correlation structure. Still, we cannot solve the Shimer puzzle this way - the volatility of unemployment is quite far from 0.125.

Figure 2: Standard deviation sensitivity to changes in ρ

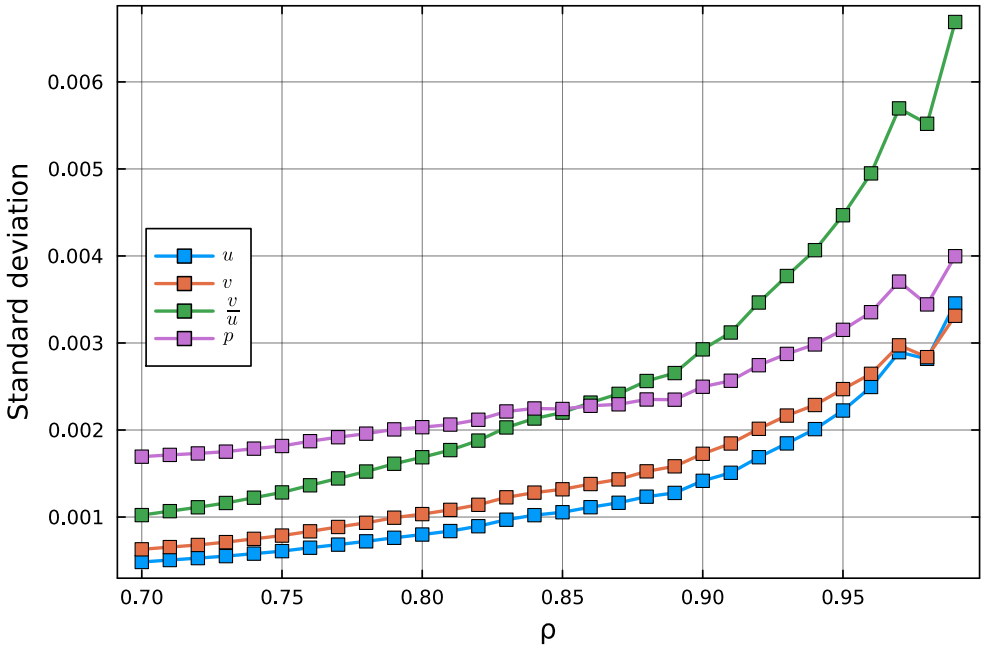
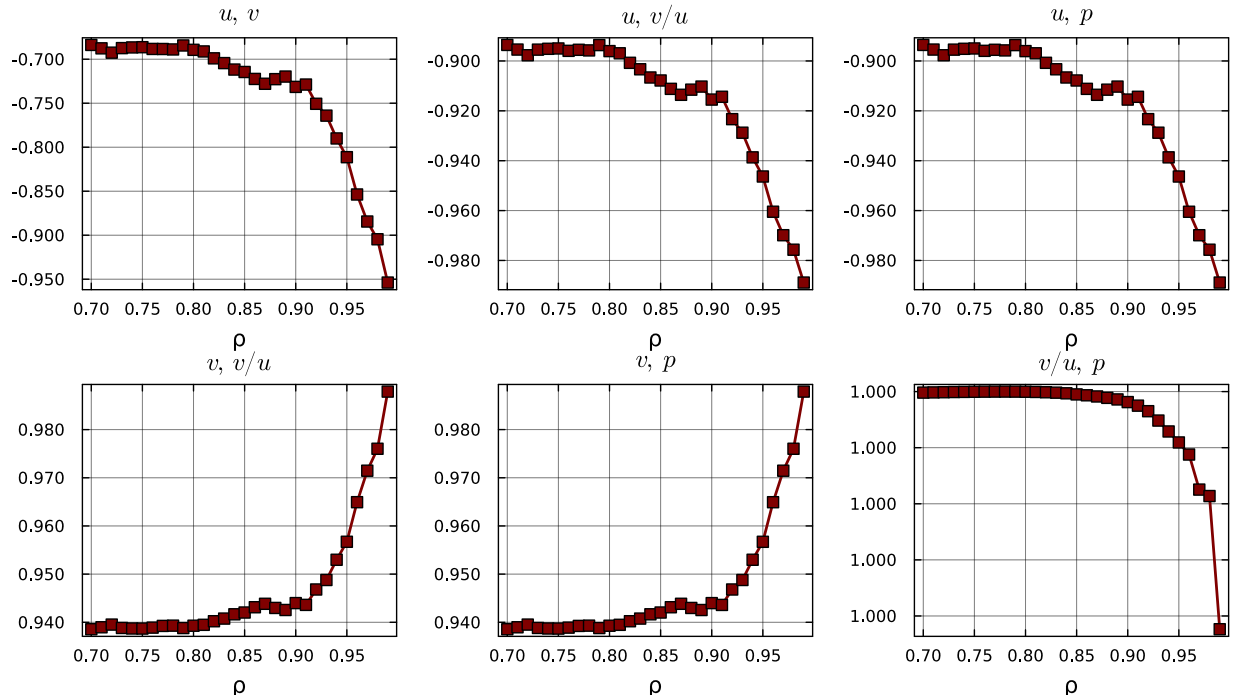


Figure 3: Correlation sensitivity to changes in ρ



At the same time, the model exhibits less sensitivity to the number of grid points (albeit very much if one is after very precise results or studying tail events). Figures 4 and 5illustrate how the key correlations

and standard deviations change for different numbers of grid points.

Figure 4: Standard deviation sensitivity to changes in N_p

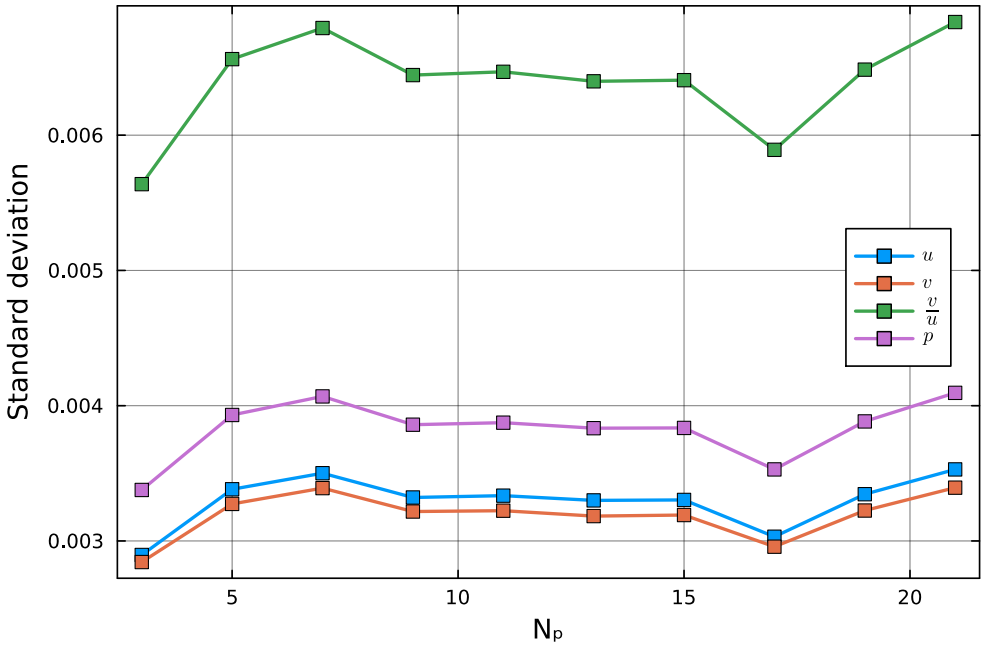
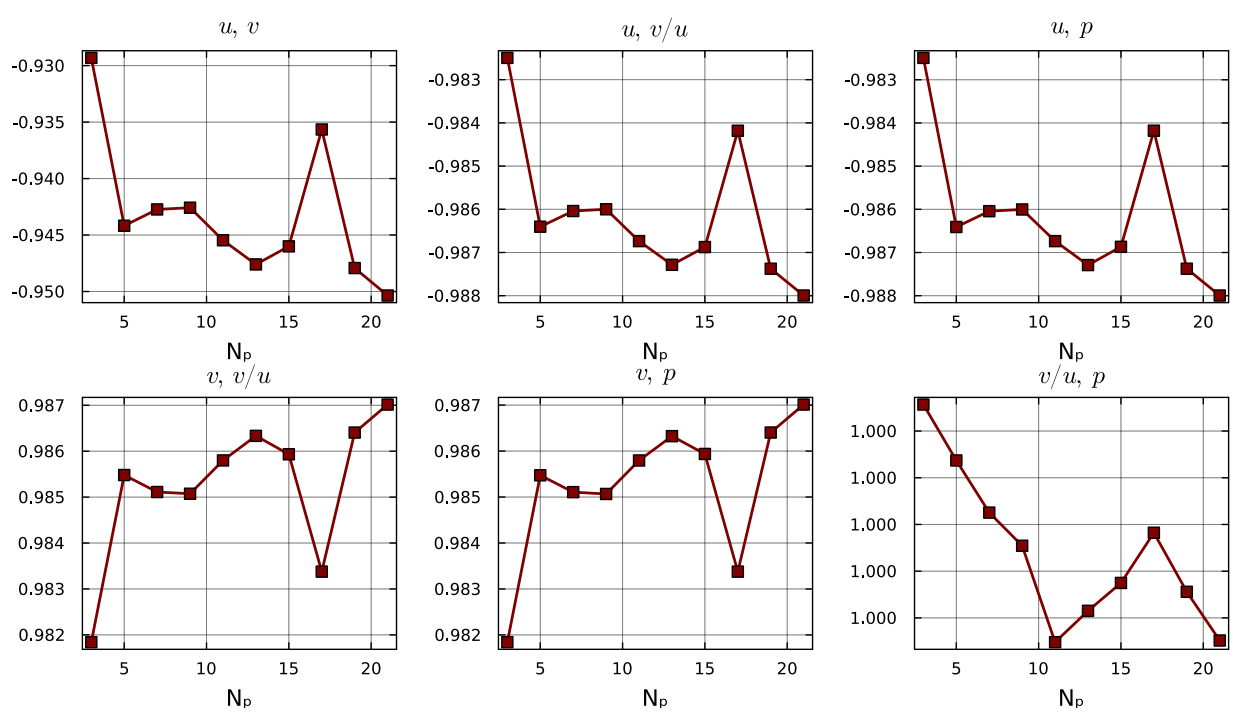


Figure 5: Correlation sensitivity to changes in N_p



Question 6: Hagedorn and Manovskii (2008)'s parameters

Table 4 features the calibrated value of c and the equilibrium outcomes under Hagedorn and Manovskii (2008)'s values of b and γ. The equilibrium wage is slightly lower, implying a moderately higher firm

value in this case. The job filling and finding rates are still the same, being pinned down by the unemployment rate which is held constant throughout the problem set.

Table 4: Steady-state calibration with alternative (b, γ)

Description	**Symbol**	**Value**
Parameters		
Home production	b	0.9400
Discount factor	β	0.9992
Separation rate	s	0.0081
Bargaining power	γ	0.0520
Matching elasticity	l	0.4070
Steady-state productivity	$p^\star$	1.0000
Calibration		
Target unemployment rate	$u^\star$	0.0550
Calibrated vacancy cost	$c^\star$	0.8161
Equilibrium objects		
Labour market tightness	$\theta^\star$	0.5997
Wage	$w^\star$	0.9686
Job-filling rate	$q^\star$	0.2321
Job-finding rate	$f^\star$	0.1392
Unemployment rate	$u^\star$	0.0550
Firm value	$J^\star$	3.5195

Table 5 redoes Question 4. Now, labour market tightness moves much more across different productivity states. Wage moves almost as little as before. What matters for us is that the job filling and finding rates vary much more, which is necessary to generate higher dispersion in unemployment.

Table 5: Equilibrium objects across productivity states under alternative (b, γ)

Description	**Symbol**	**SS**	**Min** p	**Max** p
Productivity	p	1.0000	0.9874	1.0128
Labour market tightness	θ	0.5997	0.4748	0.7315
Wage	w	0.9686	0.9626	0.9748
Job-filling rate	q	0.2321	0.2570	0.2119
Job-finding rate	f	0.1392	0.1220	0.1550
Firm value	J	3.5195	3.1750	3.8587

Table 6 redoes Question 5. I don't yet match the standard deviation from either the US data or Hagedorn and Manovskii (2008), but it's worth pointing that my productivity process is much less volatile than in their case. The firm value is not suppressed enough to generate large vacancy swings.

Table 6: Results from the calibrated model under alternative (b, γ)

	u	v	v/u	p
Standard deviation	0.034	0.033	0.066	0.004
Autocorrelation	1.000	0.999	1.000	1.000
Correlation matrix				
u	1.000	-0.945	-0.987	-0.986
v	—	1.000	0.986	0.985
v/u	—	—	1.000	0.999
p	—	—	—	1.000

Question 7: Experimentation with (b, γ)

In this question, I recompute the steady state and aggregate uncertainty versions of the model under alternative (b, γ) values. In each figure, the horizontal axis sets a different b, between 0.25 and 0.95. Then, different lines/ curves represent alternative values of γ, ranging between 0.05 and 0.95.

Figure 6's first panel illustrates different values of $c^\star$ (calibrated). The second and third panel show wages and firm values at different values b and γ. At a low bargaining power parameter and a low unemployment benefit, $c^\star$ rises and $w^\star$ falls to ensure there is still the 5.5% unemployment rate.

Figure 6: Experiment: Steady state equilibrium values

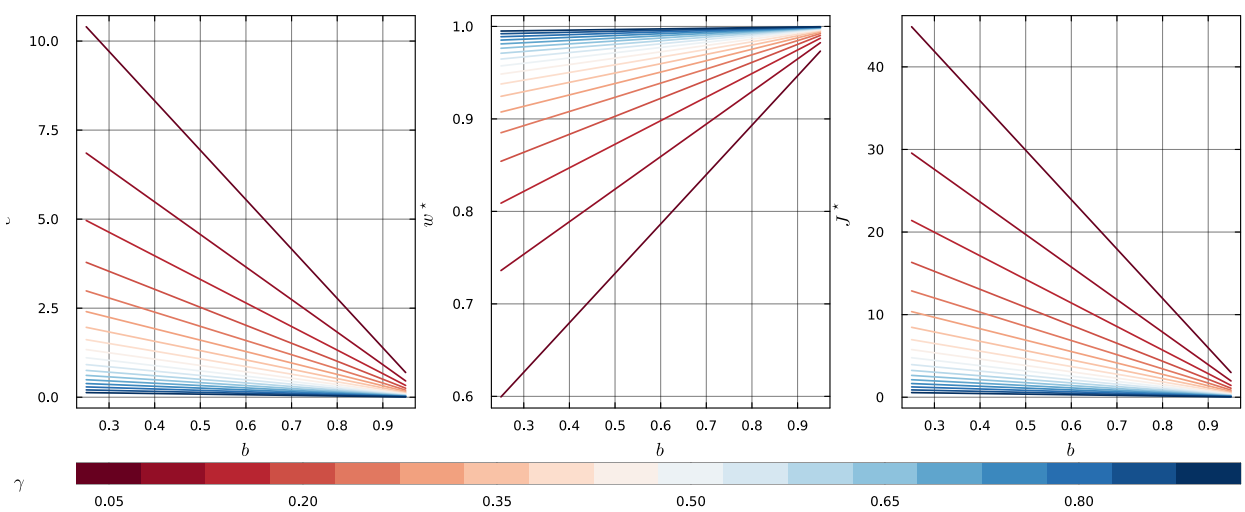
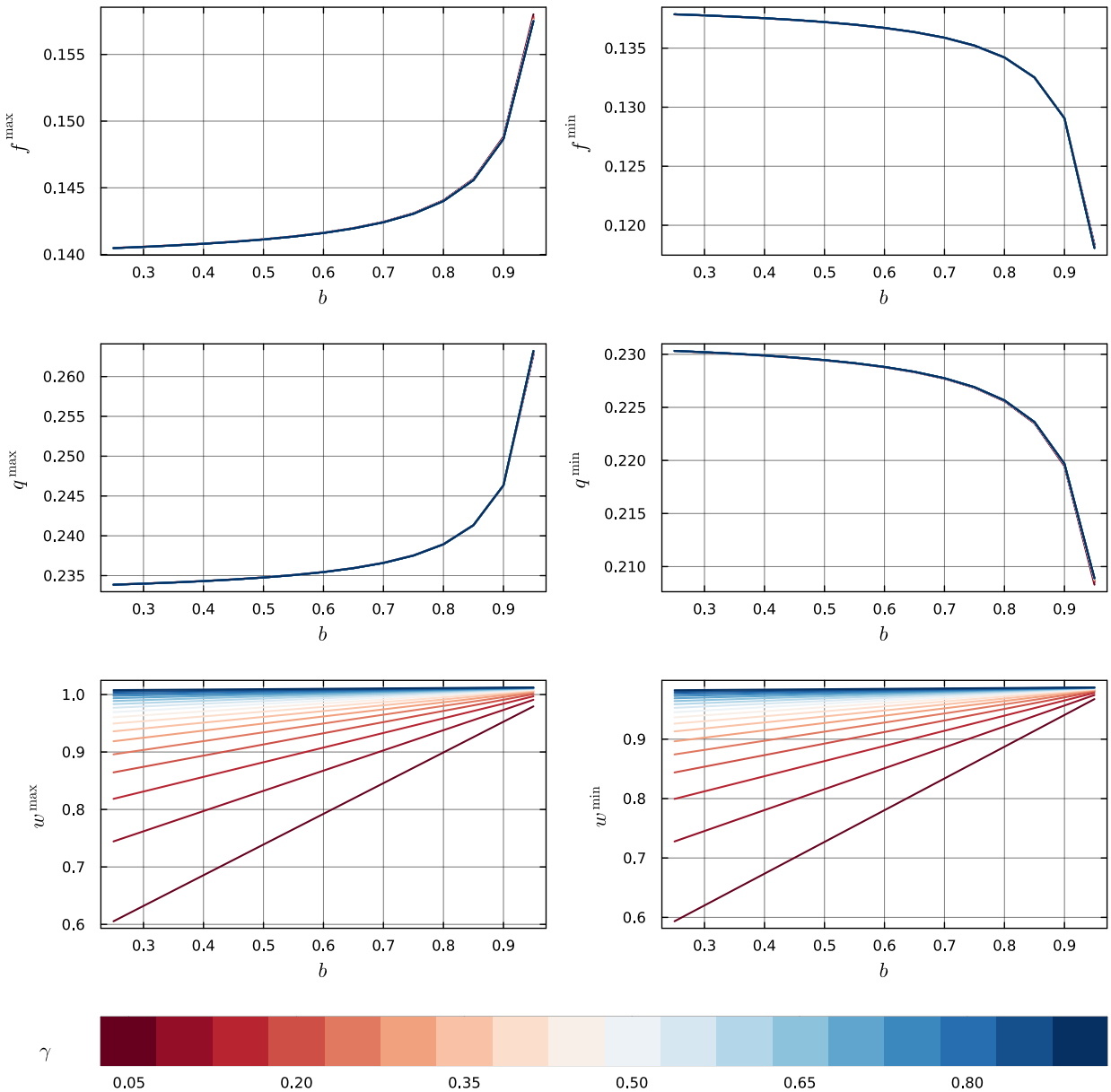


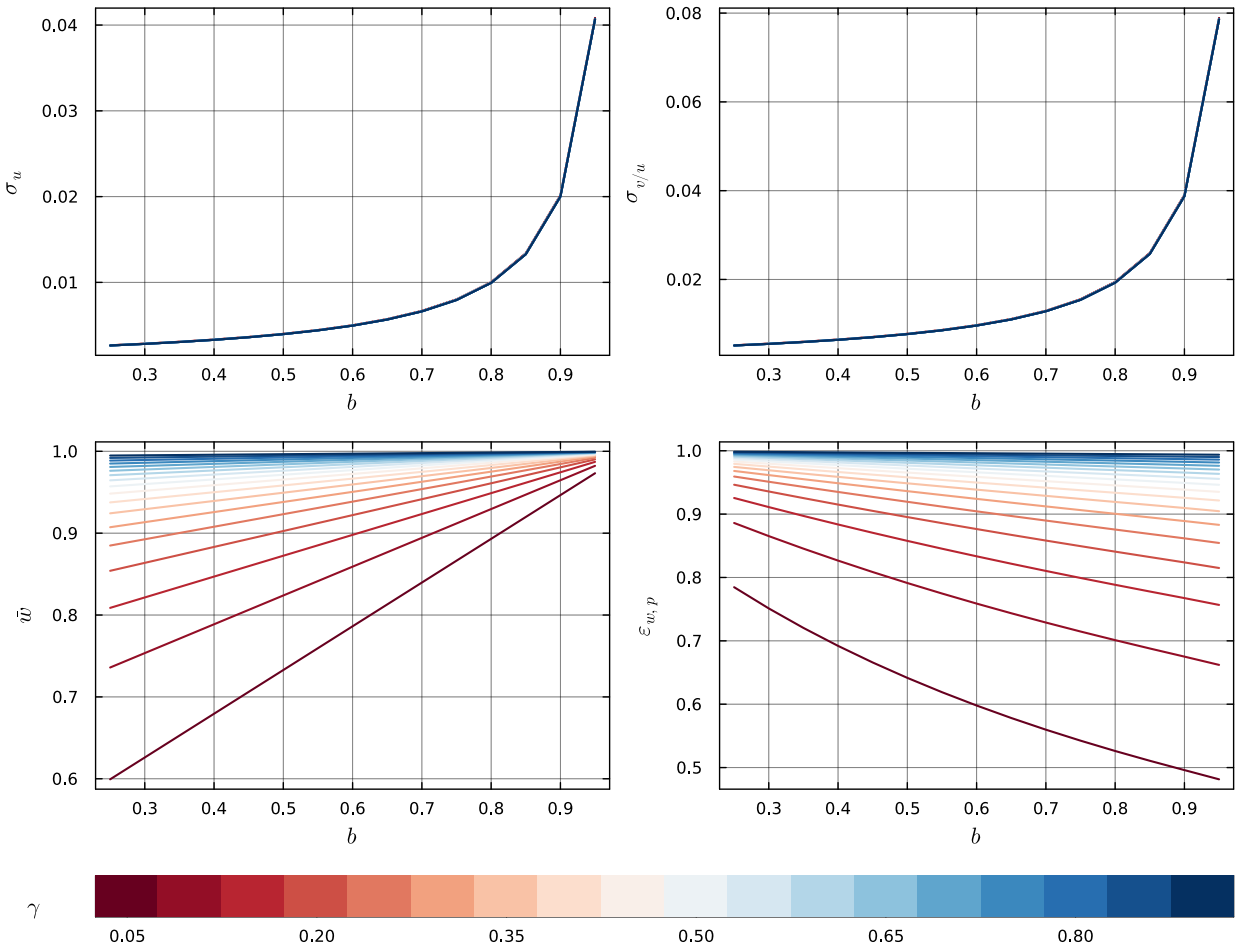
Figure 7 shows the ranges of the endogenous variables at different values of b and γ. While all of the panels clearly indicate b is dominant in impacting flows, γ matters for generating a lower wage. Given that γ's role is to "divide the pie," it does not influence the way the flow variables behave.

Figure 7: Experiment: Aggregate uncertainty ranges



Taking stock, b has a dominant impact on flows, which makes it important in determining unemployment's volatility, as illustrated in Figure 8. With a high value of b, the firm surplus is smaller, which makes it more vulnerable to swings in the productivity level. At the same time, both γ and b matter in generating the elasticity of wages with respect to the productivity process. As b becomes higher, the wage can move less – it cannot go below b, but it also cannot exceed p, thereby reducing the elasticity estimate.

Figure 8: Experiment: Simulation results



No, sticky wages are not needed to solve the Shimer puzzle/ generate an empirically realistic volatility of unemployment. A right level of b (and to a lesser extent γ) can achieve this goal. There is a deeper question, however, of whether we are satisfied by solving the puzzle by forcing the parameters to achieve the values that solve it (rather than by calibrating them to the steady state moments).

References

Den Haan, Wouter J, Garey Ramey, and Joel Watson (June 2000). "Job Destruction and Propagation of Shocks". en. In: *American Economic Review* 90.3, pp. 482–498. ISSN: 0002-8282. DOI: 10.1257/aer.90.3.482. URL: https://pubs.aeaweb.org/doi/10.1257/aer.90.3.482 (visited on 04/21/2026).

Hagedorn, Marcus and Iourii Manovskii (Aug. 2008). "The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited". en. In: *American Economic Review* 98.4, pp. 1692–1706. ISSN: 0002-8282. DOI: 10.1257/aer.98.4.1692. URL: https://pubs.aeaweb.org/doi/10.1257/aer.98.4.1692 (visited on 04/21/2026).